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V SEMESTER

B.Tech

END TERM EXAMINATION

CH301 POLYMERIC

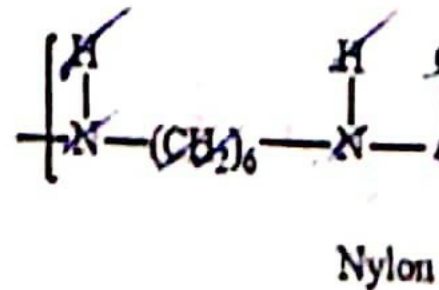
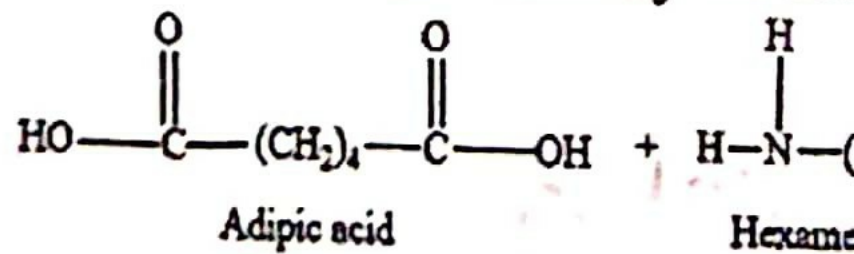
Time 3:00 Hours

Note: All questions are compulsory Assume suitable Missing data, if any
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Q.No.

- 1 Justify the statement with suitable examples.
 - [a] In thermoplastic semicrystalline polymer, crystallinity depends on rate of cooling, stereochemistry, and Polarity.
 - [b] The threshold molecular weight of polymer depends on chain length and structure.
- 2 [a] Discuss the characteristics features of

- for chain and step polymerization mechanisms.
- [b] Calculate the feed ratio of adipic acid to hexamethylenediamine that should be employed to produce a polymer with an average molecular weight of approximately 226,000 Dalton molecular weight at 100% conversion. What is the identity of the polymer?



- 3 [a] A company wants to produce a copolymer with the following monomer reactivity ratios:

M_1	r_1	M_2
Acrylonitrile	0.86	Acrylonitrile
Acrylonitrile	5.5	Vinyl chloride

Which type of copolymerization feed is used for copolymerization?

- [b] Differentiate between inhibition and retardation.

- 4 [a] Discuss the Coordination Polymerization.

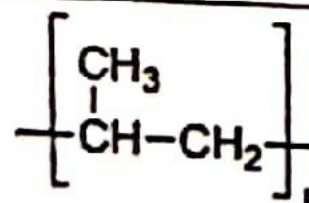
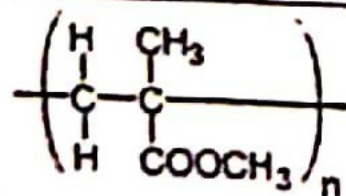
- [b] Differentiate between Redox and Anionic Polymerization.

- 5 [a] Arrange the following polymers in order of increasing glass transition temperature.

a	
b	
c	

d

[ii] Arrange the following polymers in order of increasing glass transition temperature.



a

b

[b] Differentiate the properties of amorphous polymers.

6 [a] A sample of Polyethylene is composed of different sized molecules.

Sample	Weight fraction
A	0.25
B	0.1
C	0.3
D	0.15
E	0.2

Calculate number average and weight weight and polydispersity index of this p

[b] How to determine stiffness and toughness stress- strain curve?

7 [a] The table below shows the results of a v for a sample of polystyrene in toluen degree-C. Determine the viscosity a weight of the sample, assuming the Houwink-Sakurada constants: $a = 0.63$ and

Polymer Concentration (mg/ml)	Time Flow (s)
0	76
20	219.6
16	181.4
12	150.4
8	122.1

8	122.1
4	97.4

[b] How are the mechanical properties of p temperature?

- 8 [a] How do Kelvin Voigt model and creep and stress relaxation properties of a viscoelastic polymer?
- [b] An oscillating rheometer was used to determine the mechanical properties of a viscoelastic polymer. The storage modulus and loss modulus are 1.25 MPa and 0.3834 MPa, respectively. The shear strain is 0.0125, shear stress is 0.3834°.

